ICF Practical Packet Tracer Assignment



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Task 1 - Topology

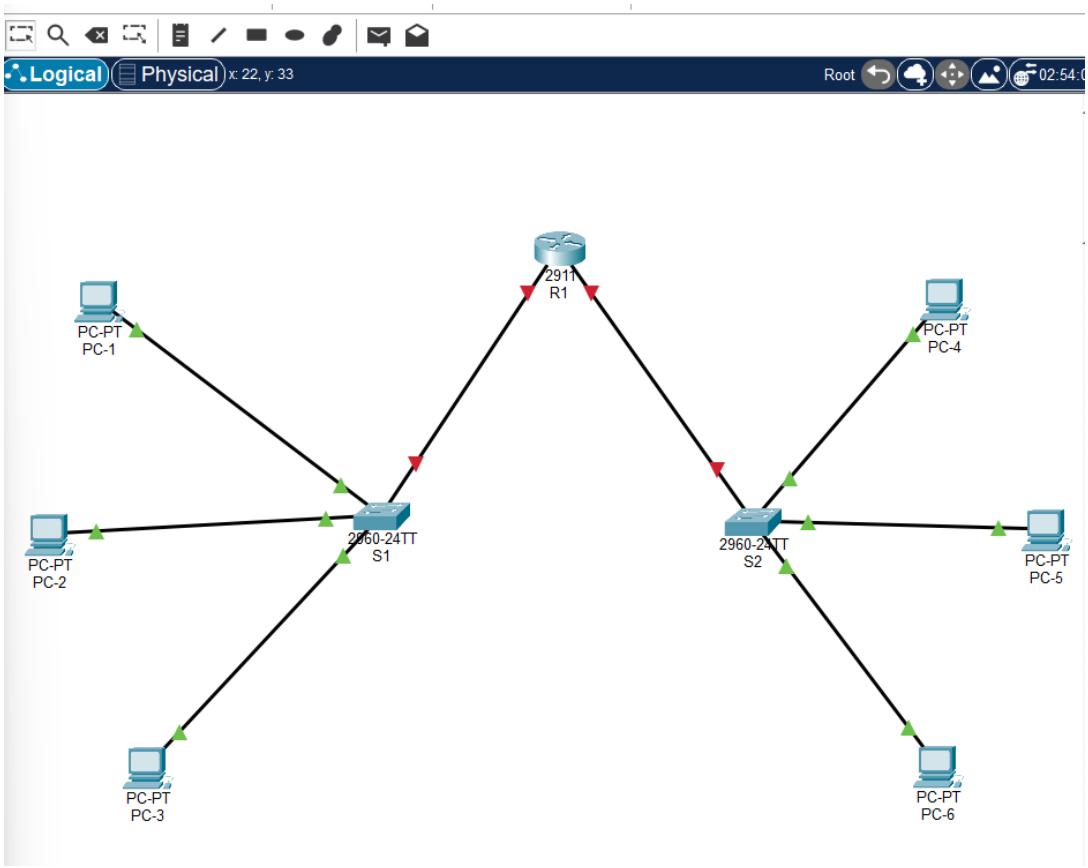


Figure 1: Completed network topology showing all devices and connection

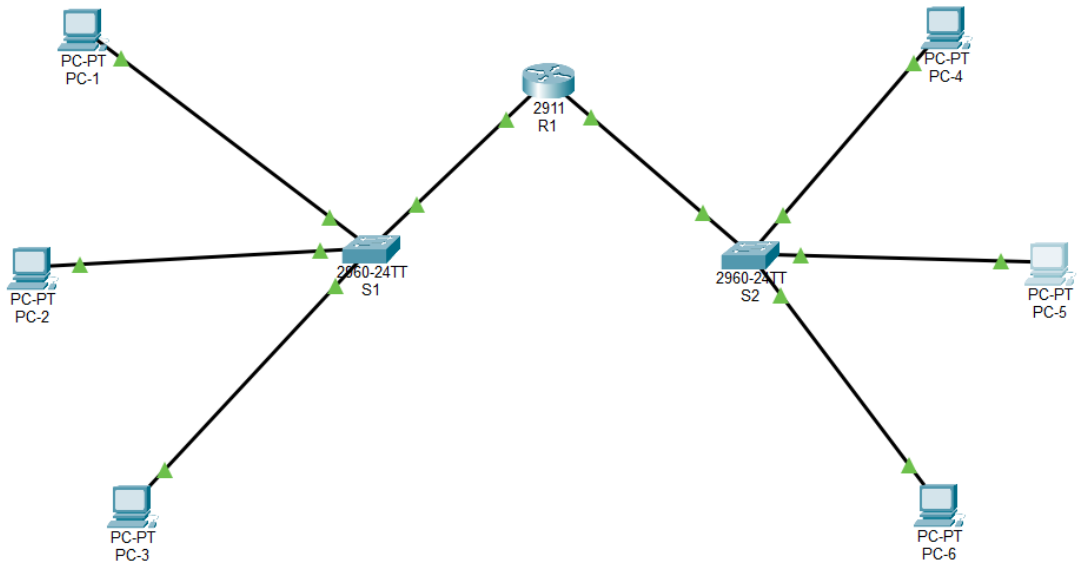


Figure 2: Completed network topology after configuration

Task 2 – Basic Router Configuration

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#no ip domain-lookup
R1(config)#enable secret class
R1(config)#line console 0
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#exit
R1(config)#service password-encryption
R1(config)#banner motd #Unauthorised access is strictly prohibited.#
```

Figure 3: R1 basic configuration including hostname, enable secret, console password, and MOTD banner

Task 3 -Router interface

```
R1(config)#interface gigabitEthernet0/0
R1(config-if)#description Link to S1 Fa0/1 (LAN 1)
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state
to up
exit
R1(config)#interface gigabitEthernet0/1
R1(config-if)#description Link to S2 Fa0/1 (LAN 2)
R1(config-if)#ip address 192.168.20.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state
to up
exit
R1(config)#show ip interface brief
           ^
% Invalid input detected at '^' marker.

R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console
show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.10.1    YES manual up                    up
GigabitEthernet0/1     192.168.20.1    YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

Figure 4: R1 interface configuration for GigabitEthernet0/0 and GigabitEthernet0/1, including 'show ip interface brief' confirming both interfaces are up/up

Task 4 – Switch S1 & S2 Configuration

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#no ip domain-lookup
S1(config)#enable secret class
S1(config)#line console 0
S1(config-line)#password cisco
S1(config-line)#login
S1(config-line)#exit
S1(config)#service password-encryption
S1(config)#banner motd #Unauthorised access is strictly prohibited.#
S1(config)#interface vlan 1
S1(config-if)#ip address 192.168.10.2 255.255.255.0
S1(config-if)#no shutdown

S1(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
exit
S1(config)#ip default-gateway 192.168.10.1
```

Figure 5: S1 configuration including hostname, console password, VLAN 1 IP, and default gateway

```
S2#
S2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
S2(config)#no ip domain-lookup
S2(config)#enable secret class
S2(config)#line console 0
S2(config-line)#password cisco
S2(config-line)#login
S2(config-line)#exit
S2(config)#service password-encryption
S2(config)#banner modt #Unauthorised acess is strictly prohibited.#
                       ^
% Invalid input detected at '^' marker.

S2(config)#banner motd #Unauthorised access is strictly prohibited.#
S2(config)#interface vlan 1
S2(config-if)#ip address 192.168.20.2 255.255.255.0
S2(config-if)#no shutdown

S2(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
exit
S2(config)#ip default-fateway 192.168.20.1
                          ^
% Invalid input detected at '^' marker.

S2(config)#ip default-gateway 192.168.20.1
S2(config)#
```

Figure 6: S2 configuration including hostname, console password, VLAN 1 IP, and default gateway

Task 5 – SSH Configuration

```
R1>enable
Password:
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip domain-name cput-lab.com
R1(config)#crypto key generate rsa modules 1024
                                   ^
% Invalid input detected at '^' marker.

R1(config)#crypto key generate rsa modulus 1024
                                   ^
% Invalid input detected at '^' marker.

R1(config)#crypto key generate rsa modulus 1024
                                   ^
% Invalid input detected at '^' marker.

R1(config)#ip ssh version 2
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
R1(config)#crypto key generate rsa modulus 1024
                                   ^
% Invalid input detected at '^' marker.

R1(config)#crypto key generate rsa
The name for the keys will be: R1.cput-lab.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#username Admin secret Adm1@55
*Mar 1 1:7:29.230: %SSH-5-ENABLED: SSH 2 has been enabled
R1(config)#line vty 0 4
R1(config-line)#transport input ssh
R1(config-line)#login local
R1(config-line)#exit
```

Figure 7: RSA key generation and SSH configuration on R1

```
S1>enable
Password:
S1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
S1(config)#ip domain-name cput-lab.com
S1(config)#ip ssh version 2
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
S1(config)#crypto key generate rsa
The name for the keys will be: S1.cput-lab.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

S1(config)#username Admin secret Adm1@55
*Mar 1 1:25:13.147: %SSH-5-ENABLED: SSH 2 has been enabled
S1(config)#line vty 0 4
S1(config-line)#transport input ssh
S1(config-line)#login local
S1(config-line)#exit
S1(config)#
```

Figure 8: RSA key generation and SSH configuration on S1

```
Unauthorised access is strictly prohibited.

User Access Verification

Password:

S2>enable
Password:
S2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
S2(config)#ip domain-name cput-lab.com
S2(config)#ip ssh version 2
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
S2(config)#crypto key generate rsa
The name for the keys will be: S2.cput-lab.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

S2(config)#username Admin secret Adm1@55
*Mar 1 1:27:27.417: %SSH-5-ENABLED: SSH 2 has been enabled
S2(config)#line vty 0 4
S2(config-line)#transport input ssh
S2(config-line)#login local
S2(config-line)#exit
S2(config)#
```

Figure 9: RSA key generation and SSH configuration on S2

Task 6 – ACL

```
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show access-lists
Standard IP access list 1
    10 deny host 192.168.20.13
    20 permit any (8 match(es))
Standard IP access list 10
    10 deny host 192.168.20.13 (4 match(es))
    20 permit any
```

Figure 10: ACL 10 applied inbound on GigabitEthernet0/1, showing 'deny host 192.168.20.13' with 4 matches and 'permit any' with 8 matches, confirming correct ACL operation

Task 7 – Verification and Testing

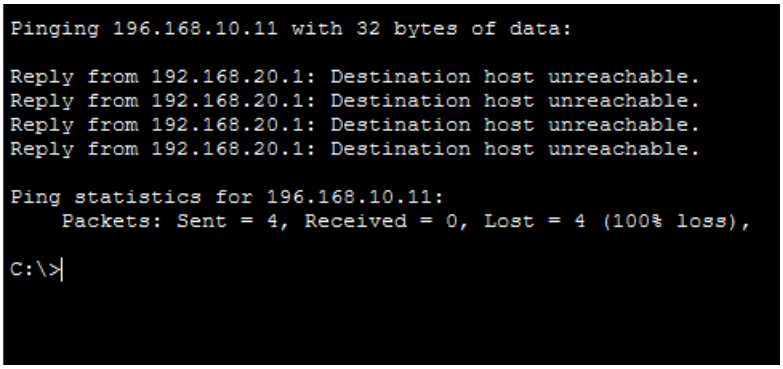


Figure 11: PC6 ping to PC1 showing "Destination host unreachable" (ACL successfully blocking PC6)

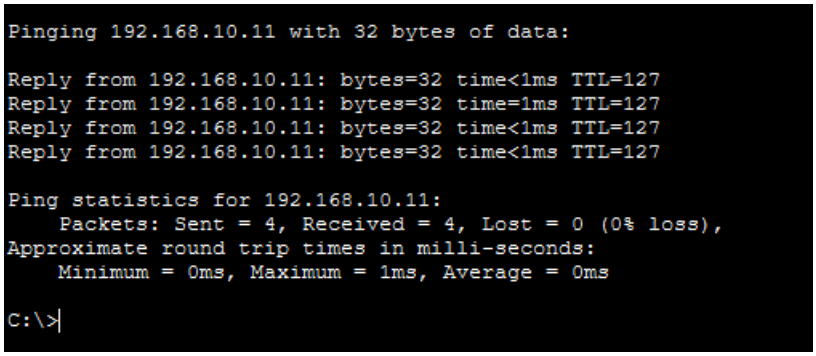


Figure 12: PC5 ping to PC1 showing successful replies (ACL allowing other hosts)

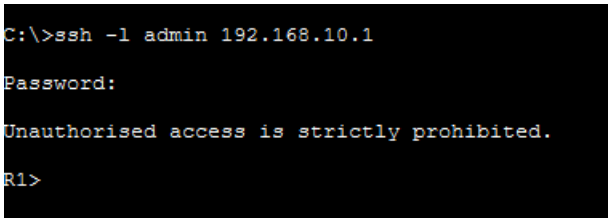


Figure 13: Successful SSH login to R1 using username Admin